

LAB 2 – Address Mapping

- Change mapping method from **page mapping** to **block mapping**

- Lab Instruction Link: [LAB2 - hackmd](#)

- Lab requirements

- 題目敘述

SimpleSSD 預設採用 page mapping。

本次作業請完成 hackmd 裡的指定的函式內容，

將其改為 block mapping。

接著請錄製一段 Demo 影片，展示程式的執行過程，並說明你實作的邏輯與設計原理。

- 範例結果

```
s98420@labtest:~/simple SSD-standalone$ ./simple SSD-standalone ./config/sample.cfg ./simple SSD/config/sample.cfg ./output
SimpleSSD Standalone v2.0
***** Begin of simulation *****
*** Statistics of Request Generator ***
Tick: 149266830923808
Time (ps): 0 - 149266830923808 (149266830923808)
I/O (bytes): 134217728 (899179.859111 B/s)
I/O (counts): 32768 (Read: 0, Write: 32768)
*** End of statistics ***
*** Statistics of Block I/O Entry ***
Latency (us): min=5.612500, max=140735472.283900, avg=145701.529996, stdev=19286.919007
*** End of statistics ***
*** Statistics of Event Engine ***
Simulation Tick (ps): 149266830923808
Host time duration (sec): 17.781976
Event handled: 102055 (5739.238430 ops)
*** End of statistics ***
End of simulation @ tick 149266830923808
```

- 評分標準

```
## Set mapping method
# Possible values:
# 0: Page level mapping
# 1: Block level mapping
MappingMode = 0
```

1 (20%) 在 [simpleSSD-standalone/simple SSD/config/config.cfg](#) 裡找到上圖的片段，將 MappingMode 改為 1，在影片中執行一次 simpleSSD，並繳交你新增的資料夾裡面的 debug.txt

```

107 ## Total size to I/O = int
108 # Only valid when time_based = false 所有IO加起來的大小
109 io_size = 128M # 128 * 2^20 bytes
110
111 ## Type of I/O = str 要產生什麼樣的IO
112 # Possible values:
113 # read: Sequential read
114 # write: Sequential write
115 # randread: Random read
116 # randwrite: Random write
117 # readwrite: Mixed sequential I/O
118 # randrw: Mixed random I/O
119 readwrite = write
120
121 ## Percentage of mixed I/O = float
122 # If readwrite is not readwrite nor randrw, this value will ignored
123 # 0 <= value <= 1 讀取在全部IO裡佔多少比例
124 rwmixread = 0.5 # 50% of read I/O

```

2 (20%) 在 [simpleSSD-standalone/config/config.cfg](#) 裡找到上圖的片段，將 readwrite 模式改為 write，io_size 改為 128 MB，在影片中執行一次 simpleSSD，並繳交你新增的資料夾裡面的 debug.txt

3 (60%) 在影片中說明程式實作的邏輯

➤ 繳交格式

hw2 /

├── code

│ ├── block_mapping.cc

│ └── 其他有更動到的程式碼檔案

├── hw2_debug_1.txt

├── hw2_debug_2.txt

└── hw2_video.mp4

➤ 繳交期限

10/27 23:59

繳交區會多開放一個禮拜至 11/3 為止，超過繳交期限就算遲交，遲交一天扣本次作業 5 分，如果到 11/3 才繳交總共會-35 分